Part I Basics

Chapter 1

Introduction

Electroencephalograms (EEGs) are ubiquitous in clinical neurology. They are used to evaluate transient neurological symptoms such as impaired awareness, altered sensations, or abnormal movements. They form a part of the evaluation of common neurological illness such as epilepsy, stroke, tumors, dementia, encephalopathy, and encephalitis. Their role in critical care medicine is increasingly being recognized. Neuroscience trainees can be sure to encounter them in the office, emergency room, at the bedside, and during various certification examinations. However, the level of comfort among trainees to confidently interpret EEGs is variable. At first, most trainees will be intimidated by their appearance and instinctively limit themselves to reading their reports. Misinterpretations of electrographic waveforms are also common resulting in needless suffering from misdiagnoses and medication misuse [1]. The best way to avoid these situations is to interpret the EEG yourself and understand its implications. Simply put, this book empowers you to do just that!

The aim of this chapter is to introduce the reader to the EEG through the following sections:

1. Basics
2. Indications
3. Limitations
4. Electrodes
5. Placing Electrodes (10–20 System)
6. Instrument
7. Display
8. Parameters
9. Calibration
10. Safety